Harder Geometrical Problems — problem 12.3.2.79

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A tetrahedron $ABCD$ is inscribed in a sphere of radius R and center O. The straight lines AO, BO, CO, DO intersect the opposite faces in A_1, B_1, C_1, D_1. Prove that

$$|AA_1| + |BB_1| + |CC_1| + |DD_1| \geq \frac{16}{3}R.$$